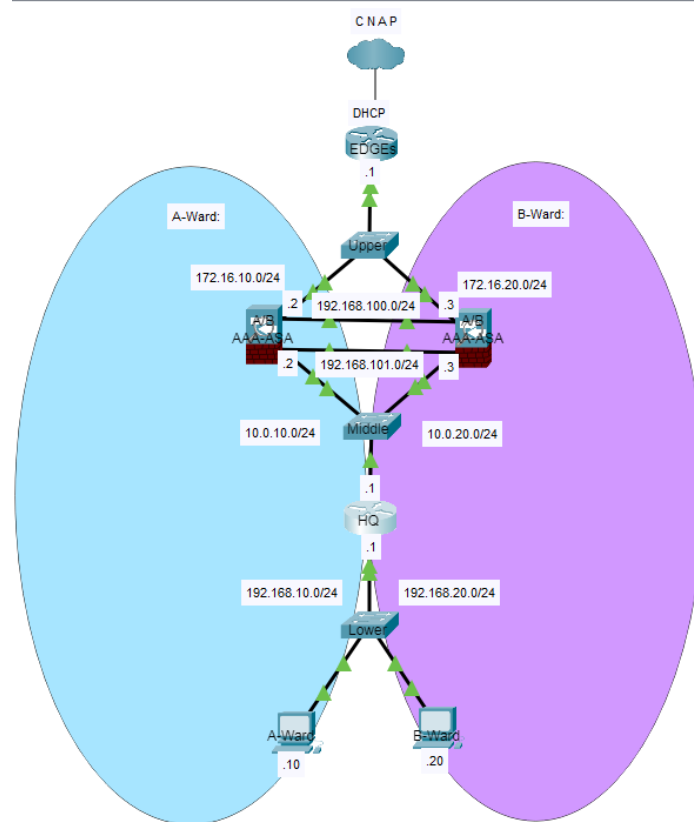


Case 4

Konfiguration av Virtualization & Failover



Topologi-bild

Grupp 9 – Pokaltagarna, Inc. (i samarbete med grupp 1)

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Sammanfattning

Två ASor har använts för att testa virtualization och redundans. Virtualization testades genom att skapa två virtuella brandväggar, oberoende av varandra, på en fysisk ASA. Tekniker som Sub-Interfaces, VRFer samt VLAN kommer användas för att separera trafiken mellan de två olika nätverken även fast de passerar samma fysiska hårdvara. Redundansen kommer testas till sin fulla potential.

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1 Inledning / Topologi

Två ASor har använts för att uppfylla redundans med hjälp av *failover*. De nätverken vi fokuserar på är *A-Ward* & *B-Ward* nätverken. Målet är att hålla trafiken separerad hela vägen fram till EDGE routern. *Figur 1, 2 & 3* visar hur routing tabellerna ser ut för respektive *EDGE*, *HQ's vrf för A-Ward*, *HQ's vrf för B-Ward*.

1.1 Internet

För senare experiment behövs anslutning till Internet. EDGE routern har kopplats upp mot CNAP-nätet och fått en adress med hjälp av DHCP. För att anslutningen ska fungera för de 2 hostarna utför routern PAT mot *192.168.10.0/24* & *192.168.20.0/24* nätverken.

2 Separation av trafik

Då separation av trafiken är ett av de primära målen har följande tekniker implementerats i topologin;

2.1 Sub-Interfaces

Sub-Interfaces användes på routern och ASorna för att göra virtuella interfaces. På routern gjordes *g0/0/1.10* & *g0/0/1.20* för respektive VLAN. Detta tillät oss att sedan använda VRFer för vardera interface. Mer om hur det användes i ASorna tas upp i 3.1.4. EDGE routern fick också Sub-Interfaces efter VLANen som implementerades på Upper-Switchen, se *figur 4*.

2.2 VRF

För att separera trafik i lager 3 implementerades VRFer på HQ routern. VRFerna har olika routing tabeller, alltså, de kan inte se varandra och är helt isolerade från varandra. Detta valdes för att till exempel använda access-listor kommer göra nätverket mer komplext än vad det behöver vara, se *figur 5*.

2.3 VLAN

VLAN användes för att separera trafik i lager 2. *Lower-switchen* har accessportar mot respektive host, se *figur 6*. Resterande portar på alla switchar mot lager 3 enheter trunkar upp VLANen som används i varje nätverk, se *figur 7, 8 & 9*. VLAN 10 & 20 används på *Lower- och Middleswitchen*. *Upperswitchen* använder VLAN 15 och 25.

3 Virtualization

Efter ni skickade förfrågan om att isolera T-101 labbet och T-800 labbet från varandra då ni var oroliga över att dem skulle hamna i konflikt så valde vi att använda oss av contexter. Efter er bifogade bild så förstod vi att det var hög prio. (se *figur 24*)

3.1 Contexts

En context är en virtuell instans som fungerar oberoende av andra contexts på samma enhet, dem är som virtuella maskiner i en hypervisor typ 2. Varje context har sin egen konfiguration. Att skapa flera contexts är särskilt användbart för er som har olika avdelningar som i ert fall är laboratorier.

3.1.1 Skapade contexter:

Dem skapade contexterna på asorna var admin, A-Ward och B-Ward

3.1.2 Konfiguration ASA:

För att skapa contexterna så behövs först användarläget på ASA:n ändras till multiple för att kunna dela in den till flera virtuella contexter. För att ändra läget användes kommandot *"mode multiple"*. Efter detta så behövs ASA:n startas om.

När *multiple* mode har aktiverats så skapas ett *"system execution space"*, att få tillgång till denna så behövs inget speciellt att göras, det räcker med att få tillgång till ASA:n med till exempel en konsol kabel eller putty. System execution space har inga nätverksinställningar utan används för att konfigurera contexter.

3.1.3 Admin context

För att kunna skapa och konfigurera contexter så behövs en admin context. För att skapa en admin context så används *"admin-context Admin"*. Själva bakgrunds konfigurationen av ett context görs med *"context {context-name}"* så med *"context Admin"* så kan vi göra konfigurationer för admin contexten. Alla contexter behöver en konfigurations fil, tilldelningen av konfigurations filer görs med *"config-url {fil-dest}"* och Admin konfigurations filen lades i disk0 med namnet admin.cfg så kommandot för detta blev *"config-url disk0:/admin.cfg"*. Att tilldela interfaces för admin contexten är inte nödvändigt men vi valde att ge admin contexten management interfacet med alias *"mange"*: *"allocate-interface Management0/0 mange."*

När admin contexten är konfigurerad så kan man börja konfigurera dem så kallade *"user context"* som i vårt fall är *"A-Ward"* och *"B-Ward"*. Skulle fler context önskas i framtiden så behövs ytterligare licenser då det endast går att ha en admin context och två user contexts på enheterna för tillfället.

3.1.4 User contexts

I System execution space så skapas ett user context med *"context {namn}"* så för A-Ward så blev det *"context A-Ward"* och motsvarande för B-Ward. Konfigurations filerna för dem båda contexterna namngavs *"A-Ward"* och *"B-Ward"* med kommandot *"config-url {namn}"*.

Då contexterna behöver interfaces tilldelade sig så behövs dem få den allokerade till sig. Då det endast fanns en fysisk kabel så fick subinterfaces användas där vlanen skickade trafik. A-Ward fick subinterfaces E0/2.15 och E0/3.10. Interfaces behöver namn så interfacet e0/2.15 fick namnet *"Aoutside"* då den är för A-Ward ut mot ISP och E0/3.10 fick namnet *"Ainside"*:

"allocate-interface Ethernet0/2.15 Aoutside".

"allocate-interface Ethernet0/3.10 Ainside".

B-Ward fick subinterfacen E0/2.25 och E0/3.20 där E0/2.25 fick namnet *"Boutside"* och E0/3.20 fick namnet *"Binside"*

3.1.5 Konfiguration i contexter

Det är i själva contexterna som man kan göra *"asa konfigurationen"* då varje context är en virtuell ASA så har dem samma konfiguration som ASA:n själv ifall den hade varit i routed mode. För att gå in i en context så används kommandot *"changeto context {namn}"*. Det är här bland annat IP adresser sätts.

Efter att *"changeto context A-Ward"* användes så var det första vi gjorde att börja konfigurera interfacen för A-Ward contexten. Då vi hade gett interfacen alias (*"Aoutside"* och *"Ainside"*) så var det dem namnen vi konfigurerade under. I *"interface Ainside"* så sattes namnet *"INSIDE"* för att få säkerhetsnivån 100 och och ip adressen 10.0.10.2/24. Aoutside fick namnet *"outside"* och ip adressen 172.16.10.2/24.

B-Wards interface "Binside" fick namnet "INSIDE" för säkerhetsnivåns skull och "Boutside" fick namnet "outside". Då B-Wards trafik inte skulle skicka till samma adresser så ändrades tredje okteten, så int Binside fick adressen 10.0.20.2/24 och Boutside fick 172.16.20.2/24.

Konfigurationen för ASA-Bs context var en kopia av ASA-A då failover användes.

4 Active/Active Failover

Active/Active innebär att båda enheterna eller systemen är aktiva i en konfiguration och de delar samtidigt på arbetsbelastningen. Varje enhet eller nod i själva systemet har sina egna uppgifter och resurser. De arbetar samtidigt för att sedan kunna distribuera belastningen och efter det kunna öka prestandan och tillgängligheten. Det finns också Active/Passive där den ena enheten är aktiv och den andra ligger på passiv och används som reserv om den aktiva enheten skulle gå ner.

En brandvägg som övervakar aktiviteten i en nätverkstrafik och tar beslut om att tillåta eller blocka överföringar baserat på kontextinformation och tillstånd av tidigare trafik kallas för en stateful firewall. Däremot har du stateless firewall också och den överväger bara enskilda paket utan att övervaka tidigare aktivitet medan en stateful firewall behåller all information om tillståndet av en anslutning, vilket kommer göra att den bättre kan fatta informerade beslut om vilken trafik som ska tillåtas eller blockas.

För att se hur det ser ut, kolla running-config och *figur 10*.

Över de två länkarna mellan ASorna användes en för control-data och en för stateful-data. Se *running-config*.

4.1 Multi-mode

Genom att vi hade två aktiva brandväggar med olika ip adresser så kunde vi öka redundansen för att om en av våra brandväggar skulle gå ner och misslyckas av någon anledning så kommer trafiken att dirigeras till den andra brandväggen i stället. Detta kommer att minimera risken för avbrott i nätverkstjänsterna. Vi ökar också tillgängligheten genom att trafiken kommer fortsätta att hanteras om brandväggen går ner och det kommer att kunna säkerställa att nätverket kommer att fortsätta fungera.

Multi-mode hjälpte oss med att distribuera trafiken jämnt mellan båda brandväggarna. Det kan vara till nytta om man tex har hög trafikbelastning. Genom att ha två brandväggar med två olika ip adresser så ökar säkerheten samt att man kan separera olika typer av trafik och tjänster.

För att se hur det ser ut, kolla running-configs och *figur 11, 12 & 13*.

4.2 SLA

Vi använde oss av SLAer då standby ASAn hade en annan IP-adress jämfört den primära. SLAn kollade *reachability* mot den primära ASAn med en frekvens på 5 sekunder. Ifall den inte skulle kunna komma åt den primära adressen kickar en floating route mot den sekundära ASAn in. *Figur 14, 15 & 16* visar hur SLAerna sattes upp.

Efter vi använde SLAer och det inte fungerade märkte vi att vi behövde *monitorera* ändringar på interfacarna. Se *figur 17, 18, 19, 20 & 21*.

5 Real-time Failover Demonstration

Lite extra tester;

5.1 Single-mode

För att få en utökad ökad förståelse av hur Failover fungerar så genomfördes ett separat test på det i "Single mode" dvs utan flera contexts. Samma topologi användes men ingen separation av trafiken med VLAN är implementerad.

Under det här testet så insågs att IP adresserna flyttas/skiftas emellan ASA enheterna så att den som är "Active" alltid har samma "Active" adress. Som en förklaring så kommer ASA:n som är "Active" alltid ha samma adress och den ASA:n som är i "Standby" kommer alltid att ha samma adress oavsett vilken av ASA enheterna det är. Om "Active" ASA:n slutar fungera så skiftas adresserna emellan ASA enheterna så att den som är "Active" alltid har samma adress. Se *figur 22* och i *figur 23* för en demonstration av detta.

I *Figur 22* så är "Primary" ASA:n "Active" och har IP adressen 10.0.0.2 för sitt "OUTSIDE" interface och IP adressen 192.168.1.2 för sitt "INSIDE" interface medans "Secondary" ASA:n är i "Standby" och har IP adressen 10.0.0.3 för sitt "OUTSIDE" interface och IP adressen 192.168.1.3 för sitt "INSIDE" interface. Om man skiftar blicken till *figur 23* så ser man att en failover har skett och att "Standby" ASA:n nu är "Active" och att adresserna har skiftats så att denna nu har 10.0.0.2 och 192.168.1.2.

Detta betyder att i vår lösning så hade det inte behövts SLA:er eftersom adressen inte ändras till standby adressen som vi först trodde. All trafik åker fortfarande mot samma adress bara att den är på en den andra enheten.

5.2 HALO!!! :D

För att visa hur snabbt och effektivt failover fungerar i realtid så genomfördes en praktiskt test / demonstration av detta med hjälp av ett Xbox och spelet Halo. Själva testet utgick på att spela mutliplayer matcher i Halo och sedan dra ut en kabel från den ASA:n som är "Active" för att trigga en failover. Under en multiplayer match i Halo, om spelet förlorar kontakt med dess servrar för länge så blir man borttagen från matchen och får en temporär ban för att man har, i deras ögon, lämnat matchen medan den var igång. I demonstrationen som syns i videon länkad nedan så dras en kabel ut och spelet laggar till lite eller får en mycket kort laddningsskärm när själva failovern sker. Eftersom vi använder oss av statfull failover så skickas all information om alla aktiva kopplingar över till den andra ASA:n.

Länk till realtids Demonstration av failover:

https://www.youtube.com/watch?v=4Q3UTBwkbNs&list=PLThJUQZOLMG7nZyAg0Y-DDMiUwvDaec7q&ab_channel=NetworkDuck

6 Bilagor

```

+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is 193.10.203.129 to network 0.0.0.0

S*   0.0.0.0/0 [254/0] via 193.10.203.129
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     10.0.0.0/24 is directly connected, GigabitEthernet0/0/0
L     10.0.0.1/32 is directly connected, GigabitEthernet0/0/0
     172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C     172.16.10.0/24 is directly connected, GigabitEthernet0/0/0.15
L     172.16.10.1/32 is directly connected, GigabitEthernet0/0/0.15
C     172.16.20.0/24 is directly connected, GigabitEthernet0/0/0.25
L     172.16.20.1/32 is directly connected, GigabitEthernet0/0/0.25
S     192.168.0.0/16 [1/0] via 172.16.10.2
S     192.168.10.0/24 [1/0] via 172.16.10.2
S     192.168.20.0/24 [1/0] via 172.16.20.2
     193.10.203.0/24 is variably subnetted, 2 subnets, 2 masks
C     193.10.203.128/25 is directly connected, GigabitEthernet0/0/1
L     193.10.203.150/32 is directly connected, GigabitEthernet0/0/1
```

Figur 1 – EDGEs routing tabell

```

HQ#show ip route vrf A-Ward

Routing Table: A-Ward
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 10.0.10.2 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 10.0.10.2
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     10.0.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L     10.0.10.1/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.10.0/24 is directly connected, GigabitEthernet0/0/1.10
L     192.168.10.1/32 is directly connected, GigabitEthernet0/0/1.10
```

Figur 2 – HQ's vrf för A-Ward

```

Gateway of last resort is 10.0.20.2 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 10.0.20.2
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     10.0.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L     10.0.20.1/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.20.0/24 is directly connected, GigabitEthernet0/0/1.20
L     192.168.20.1/32 is directly connected, GigabitEthernet0/0/1.20
```

Figur 3 – HQ's vrf för B-Ward

```

EDGEs#show ip inte brief
Interface                               IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0                   10.0.0.1        YES manual  up          up
Gi0/0/0.15                             172.16.10.1     YES manual  up          up
Gi0/0/0.25                             172.16.20.1     YES manual  up          up
GigabitEthernet0/0/1                   193.10.203.150  YES DHCP    up          up
Serial0/1/0                            unassigned      YES unset   down        down
Serial0/1/1                            unassigned      YES unset   down        down
Serial0/2/0                            unassigned      YES unset   down        down
Serial0/2/1                            unassigned      YES unset   down        down
GigabitEthernet0                       unassigned      YES unset   administratively down down
```

Figur 4 – EDGEs Sub-Interfaces


```

HQ(config)#do show vrf

```

Name	Default RD	Protocols	Interfaces
A-Ward	<not set>	ipv4,ipv6	Gi0/0/0.10 Gi0/0/1.10
B-Ward	<not set>	ipv4	Gi0/0/0.20 Gi0/0/1.20
Mgmt-intf	<not set>	ipv4,ipv6	Gi0

Figur 5 – HQ's VRFs

```

Lower-Switch#show vlan

```

VLAN Name	Status	Ports
1 default	active	Gi1/0/3, Gi1/0/4, Gi1/0/5 Gi1/0/6, Gi1/0/7, Gi1/0/8 Gi1/0/9, Gi1/0/10, Gi1/0/11 Gi1/0/12, Gi1/0/13, Gi1/0/14 Gi1/0/15, Gi1/0/16, Gi1/0/17 Gi1/0/18, Gi1/0/19, Gi1/0/20 Gi1/0/21, Gi1/0/22, Gi1/0/23 Gi1/0/25, Gi1/0/26, Gi1/0/27 Gi1/0/28, Gi1/0/29, Gi1/0/30 Gi1/0/31, Gi1/0/32, Gi1/0/33 Gi1/0/34, Gi1/0/35, Gi1/0/36 Gi1/0/37, Gi1/0/38, Gi1/0/39 Gi1/0/40, Gi1/0/41, Gi1/0/42 Gi1/0/43, Gi1/0/44, Gi1/0/45 Gi1/0/46, Gi1/0/47, Gi1/0/48 Gi1/0/49, Gi1/0/50, Gi1/0/51 Gi1/0/52
10 A-Ward	active	Gi1/0/1
20 B-Ward	active	Gi1/0/2
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

Figur 6 – Lower-Switch VLAN

```

Middle-Switch#show int trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Gi1/0/1	on	802.1q	trunking	1
Gi1/0/2	on	802.1q	trunking	1
Gi1/0/24	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gi1/0/1	10,20
Gi1/0/2	10,20
Gi1/0/24	10,20

Port	Vlans allowed and active in management domain
Gi1/0/1	10,20
Gi1/0/2	10,20
Gi1/0/24	10,20

Port	Vlans in spanning tree forwarding state and not pruned
Gi1/0/1	10,20
Gi1/0/2	10,20
Gi1/0/24	10,20

Figur 7 – Middle-Switch trunks

```
Lower-Switch#show int trunk

Port      Mode      Encapsulation  Status      Native vlan
Gi1/0/24  on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gi1/0/24  10,20

Port      Vlans allowed and active in management domain
Gi1/0/24  10,20

Port      Vlans in spanning tree forwarding state and not pruned
Gi1/0/24  10,20
```

Figur 8 – Lower-Switch trunks

```
Upper-Switch(config)#do show int trunk

Port          Mode          Encapsulation  Status        Native vlan
Gi1/0/1       on            802.1q         trunking      1
Gi1/0/2       on            802.1q         trunking      1
Gi1/0/24      on            802.1q         trunking      1

Port          Vlans allowed on trunk
Gi1/0/1       15,25
Gi1/0/2       15,25
Gi1/0/24      15,25

Port          Vlans allowed and active in management domain
Gi1/0/1       15,25
Gi1/0/2       15,25
Gi1/0/24      15,25

Port          Vlans in spanning tree forwarding state and not pruned
Gi1/0/1       15,25
Gi1/0/2       15,25
Gi1/0/24      15,25
Upper-Switch(config)#
```

Figur 9 – Upper-Switch trunks

```

Group 1 last failover at: 05:39:35 UTC Jan 2 2003
Group 2 last failover at: 05:35:30 UTC Jan 2 2003

This host: Primary
Group 1 State: Failed
Active time: 8265 (sec)
Group 2 State: Failed
Active time: 8139 (sec)

slot 0: ASA5510 hw/sw rev (2.0/9.0(3)) status (Up Sys)
Admin Interface Mangle (0.0.0.0): No Link (Waiting)
A-Ward Interface INSIDE (10.0.10.3): No Link (Waiting)
B-Ward Interface OUTSIDE (172.16.10.3): Normal (Waiting)
B-Ward Interface outside (172.16.20.3): Normal (Monitored)
B-Ward Interface inside (10.0.20.3): No Link (Monitored)
slot 1: ASA-38M-10 hw/sw rev (1.0/7.0(4)E4) status (Up/Up)
IPS, 7.0(4)E4, Up

Other host: Secondary
Group 1 State: Active
Active time: 32 (sec)
Group 2 State: Active
Active time: 278 (sec)

slot 0: ASA5510 hw/sw rev (2.0/9.1(5)) status (Up Sys)
Admin Interface Mangle (192.168.99.1): No Link (Waiting)
A-Ward Interface INSIDE (10.0.10.2): Normal (Waiting)
A-Ward Interface OUTSIDE (172.16.10.2): Normal (Waiting)
B-Ward Interface outside (172.16.20.2): Normal (Monitored)
B-Ward Interface inside (10.0.20.2): Normal (Monitored)
slot 1: ASA-38M-10 hw/sw rev (1.0/) status (Unresponsive/Up)

Stateful Failover Logical Update Statistics
Link : FailoverState Ethernet0/1 (up)
Stateful Obj xmit xerr rcv rerr
General 6187 0 1460 0
sys cmd 1242 0 1241 0
up time 0 0 0 0
RPC services 0 0 0 0
TCP conn 2598 0 131 0

```

Figur 10 – Failover test

```

ASA-AAA/A-Ward# show failover
Failover On
Last Failover at: 04:51:44 UTC Jan 2 2003
  This context: Active
    Active time: 5715 (sec)
    Interface INSIDE (10.0.10.2): Normal (Monitored)
    Interface OUTSIDE (172.16.10.2): Normal (Monitored)
  Peer context: Standby Ready
    Active time: 111 (sec)
    Interface INSIDE (10.0.10.3): Normal (Monitored)
    Interface OUTSIDE (172.16.10.3): Normal (Monitored)

Stateful Failover Logical Update Statistics
Status: Configured.
Stateful Obj  xmit      xerr      rcv        rerr
RPC services   0          0          0          0
TCP conn      1582       0          61         0
UDP conn       780       0          26         0
ARP tbl       11         0          8          0
Xlate Timeout  0          0          0          0
IPv6 ND tbl    0          0          0          0
VPN IKEv1 SA   0          0          0          0
VPN IKEv1 P2   0          0          0          0
VPN IKEv2 SA   0          0          0          0
VPN IKEv2 P2   0          0          0          0
VPN CTCP upd   0          0          0          0
VPN SDI upd    0          0          0          0
VPN DHCP upd   0          0          0          0
SIP Session    0          0          0          0
Route Session  0          0          0          0
User-Identity  3          0          0          0
CTS SGTNAME    0          0          0          0
CTS PAC        0          0          0          0
TrustSec-SXP   0          0          0          0
IPv6 Route     0          0          0          0

```

Figur 11 – A-Ward show failover

```

Failover On
Last Failover at: 04:51:44 UTC Jan 2 2003
  This context: Standby Ready
    Active time: 111 (sec)
    Interface INSIDE (10.0.10.3): Normal (Monitored)
    Interface OUTSIDE (172.16.10.3): Normal (Monitored)
  Peer context: Active
    Active time: 6744 (sec)
    Interface INSIDE (10.0.10.2): Normal (Monitored)
    Interface OUTSIDE (172.16.10.2): Normal (Monitored)

Stateful Failover Logical Update Statistics
Status: Configured.
Stateful Obj  xmit      xerr      rcv        rerr
RPC services   0          0          0          0
TCP conn       61         0        1399        0
UDP conn       27         0        494         0
ARP tbl        8         0          3          0
Xlate Timeout  0          0          0          0
IPv6 ND tbl    0          0          0          0
VPN IKEv1 SA   0          0          0          0
VPN IKEv1 P2   0          0          0          0
VPN IKEv2 SA   0          0          0          0
VPN IKEv2 P2   0          0          0          0
VPN CTCP upd   0          0          0          0
VPN SDI upd    0          0          0          0
VPN DHCP upd   0          0          0          0
SIP Session    0          0          0          0
Route Session  0          0          0          0
User-Identity  0          0          1          0
CTS SGTNAME    0          0          0          0
CTS PAC        0          0          0          0
TrustSec-SXP   0          0          0          0
IPv6 Route     0          0          0          0

```

Figur 12 – A-Ward standby

```

ASA-AAA/B-Ward# show failover
Failover On
Last Failover at: 04:51:44 UTC Jan 2 2003
  This context: Active
    Active time: 5965 (sec)
    Interface outside (172.16.20.2): Normal (Monitored)
    Interface inside (10.0.20.2): Normal (Monitored)
  Peer context: Standby Ready
    Active time: 329 (sec)
    Interface outside (172.16.20.3): Normal (Monitored)
    Interface inside (10.0.20.3): Normal (Monitored)

Stateful Failover Logical Update Statistics
Status: Configured.
Stateful Obj  xmit      xerr      rcv        rerr
RPC services  0          0          0           0
TCP conn      176        0          19          0
UDP conn      1135       0          10          0
ARP tbl       9          0          8           0
Xlate Timeout 0          0          0           0
IPv6 ND tbl   0          0          0           0
VPN IKEv1 SA  0          0          0           0
VPN IKEv1 P2  0          0          0           0
VPN IKEv2 SA  0          0          0           0
VPN IKEv2 P2  0          0          0           0
VPN CTCP upd  0          0          0           0
VPN SDI upd   0          0          0           0
VPN DHCP upd  0          0          0           0
SIP Session   0          0          0           0
Route Session 0          0          0           0
User-Identity 3          0          0           0
CTS SGTNAME   0          0          0           0
CTS PAC       0          0          0           0
TrustSec-SXP  0          0          0           0
IPv6 Route    0          0          0           0
ASA-AAA/B-Ward#

```

Figur 13 – B-Ward show failover

```

EDGEs#show ip sla summ
IPSLAs Latest Operation Summary
Codes: * active, ^ inactive, ~ pending
All Stats are in milliseconds. Stats with u are in microseconds

ID          Type          Destination      Stats          Return      Last
-----
*4          icmp-echo      172.16.10.2     RTT=1          OK           1 seconds ago

*8          icmp-echo      172.16.20.2     RTT=1          OK           1 seconds ago

```

Figur 14 – EDGE SLA summary

```

HQ#show ip sla summ
IPSLAs Latest Operation Summary
Codes: * active, ^ inactive, ~ pending
All Stats are in milliseconds. Stats with u are in microseconds

ID          Type          Destination      Stats          Return      Last
-----
*4          icmp-echo      10.0.10.2       RTT=1          OK           2 seconds ago

*8          icmp-echo      10.0.20.2       RTT=1          OK           1 seconds ago

```

Figur 15 – HQ SLA summary


```

HQ#show track
Track 4
  IP SLA 4 reachability
  Reachability is Up
    9 changes, last change 00:37:16
  Delay up 15 secs, down 10 secs
  Latest operation return code: OK
  Latest RTT (milliseconds) 1
  Tracked by:
    Static IP Routing 0
Track 8
  IP SLA 8 reachability
  Reachability is Up
    9 changes, last change 00:37:16
  Delay up 15 secs, down 10 secs
  Latest operation return code: OK
  Latest RTT (milliseconds) 1
  Tracked by:
    Static IP Routing 0

```

Figur 16 – HQ Tracks

```

ASA-AAA/B-Ward(config)# monitor-interface inside
ASA-AAA/B-Ward(config)# monitor-interface out
ASA-AAA/B-Ward(config)# monitor-interface outside
ASA-AAA/B-Ward(config)# exit
ASA-AAA/B-Ward# changeto context B-Ward
ASA-AAA/B-Ward# mon
ASA-AAA/B-Ward# moni
ASA-AAA/B-Ward# conf t
ASA-AAA/B-Ward(config)# exit
ASA-AAA/B-Ward# changeto context A-Ward

```

Figur 17 – Enabeling monitoring

```

ASA-AAA/A-Ward# show monitor
  This host: Primary - Active
    Interface INSIDE (10.0.10.2): Normal (Monitored)
    Interface OUTSIDE (172.16.10.2): Normal (Monitored)
  Other host: Secondary - Standby Ready
    Interface INSIDE (10.0.10.3): Normal (Monitored)
    Interface OUTSIDE (172.16.10.3): Normal (Monitored)
ASA-AAA/A-Ward#

```

Figur 18 – Show monitor A-Ward

```

Failover On
Last Failover at: 04:51:44 UTC Jan 2 2003
  This context: Standby Ready
    Active time: 111 (sec)
    Interface INSIDE (10.0.10.3): Normal (Monitored)
    Interface OUTSIDE (172.16.10.3): Normal (Monitored)
  Peer context: Active
    Active time: 6744 (sec)
    Interface INSIDE (10.0.10.2): Normal (Monitored)
    Interface OUTSIDE (172.16.10.2): Normal (Monitored)

Stateful Failover Logical Update Statistics
Status: Configured.
Stateful Obj   xmit      xerr      rcv        rerr
RPC services   0          0          0           0
TCP conn       61         0         1399        0
UDP conn       27         0          494        0
ARP tbl        8          0           3           0
Xlate_Timeout  0          0           0           0
IPv6 ND tbl    0          0           0           0
VPN IKEv1 SA   0          0           0           0
VPN IKEv1 P2   0          0           0           0
VPN IKEv2 SA   0          0           0           0
VPN IKEv2 P2   0          0           0           0
VPN CTCP upd   0          0           0           0
VPN SDI upd    0          0           0           0
VPN DHCP upd   0          0           0           0
SIP Session    0          0           0           0
Route Session  0          0           0           0
User-Identity  0          0           1           0
CTS SGTNAME    0          0           0           0
CTS PAC        0          0           0           0
TrustSec-SXP   0          0           0           0
IPv6 Route     0          0           0           0
ASA-AAA/A-Ward(config)# show monitor
  This host: Secondary - Standby Ready
    Interface INSIDE (10.0.10.3): Normal (Monitored)
    Interface OUTSIDE (172.16.10.3): Normal (Monitored)
  Other host: Primary - Active
    Interface INSIDE (10.0.10.2): Normal (Monitored)
    Interface OUTSIDE (172.16.10.2): Normal (Monitored)

```

Figur 19 – Show monitor A-Ward, 2nd

```

ASA-AAA/B-Ward# show monitor
  This host: Primary - Active
    Interface outside (172.16.20.2): Normal (Monitored)
    Interface inside (10.0.20.2): Normal (Monitored)
  Other host: Secondary - Standby Ready
    Interface outside (172.16.20.3): Normal (Monitored)
    Interface inside (10.0.20.3): Normal (Monitored)

```

Figur 20 – Show monitor B-Ward

```

ASA-AAA/B-Ward(config)# show monitor
  This host: Secondary - Standby Ready
    Interface inside (10.0.20.3): Normal (Monitored)
    Interface outside (172.16.20.3): Normal (Monitored)
  Other host: Primary - Active
    Interface inside (10.0.20.2): Normal (Monitored)
    Interface outside (172.16.20.2): Normal (Monitored)

```

Figur 21 – Show monitor B-Ward, 2nd

```

Unit Poll frequency 1 seconds, holdtime 15 seconds
Interface Poll frequency 5 seconds, holdtime 25 seconds
Interface Policy 1
Monitored Interfaces 2 of 110 maximum
failover replication http
Version: Ours 9.0(3), Mate 9.0(3)
Last Failover at: 01:37:16 UTC Jan 1 2003
  This host: Primary - Active
    Active time: 3284 (sec)
    slot 0: ASA5510 hw/sw rev (2.0/9.0(3)) status (Up Sys)
      Interface OUTSIDE (10.0.0.2): Normal (Waiting)
      Interface INSIDE (192.168.1.2): Normal (Waiting)
    slot 1: empty
  Other host: Secondary - Standby Ready
    Active time: 640 (sec)
    slot 0: ASA5510 hw/sw rev (2.0/9.0(3)) status (Up Sys)
      Interface OUTSIDE (10.0.0.3): Normal (Waiting)
      Interface INSIDE (192.168.1.3): Normal (Waiting)
    slot 1: empty

```

Figur 22 - Primary

```

Halo-Bois(config)# sh failover
Failover On
Failover unit Primary
Failover LAN Interface: FailoverControl Ethernet0/0 (up)
Unit Poll frequency 1 seconds, holdtime 15 seconds
Interface Poll frequency 5 seconds, holdtime 25 seconds
Interface Policy 1
Monitored Interfaces 2 of 110 maximum
failover replication http
Version: Ours 9.0(3), Mate 9.0(3)
Last Failover at: 01:16:19 UTC Jan 1 2003
  This host: Primary - Standby Ready
    Active time: 2592 (sec)
    slot 0: ASA5510 hw/sw rev (2.0/9.0(3)) status (Up Sys)
      Interface OUTSIDE (10.0.0.3): Normal (Monitored)
      Interface INSIDE (192.168.1.3): Failed (Waiting)
    slot 1: empty
  Other host: Secondary - Active
    Active time: 115 (sec)
    slot 0: ASA5510 hw/sw rev (2.0/9.0(3)) status (Up Sys)
      Interface OUTSIDE (10.0.0.2): Normal (Monitored)
      Interface INSIDE (192.168.1.2): Normal (Waiting)
    slot 1: empty

```

Figur 23 - Secondary



Figur 24 - 3rr0r

7 Running-configs

Se nedanför :D

7.1 Admin primary

: Saved

:

ASA Version 9.0(3) <context>

!

hostname Admin

enable password 8Ry2YjlYt7RRXU24 encrypted

names

!

interface Mange

management-only

nameif Mange

security-level 100

ip address 192.168.99.1 255.255.255.0

!

pager lines 24

mtu Mange 1500

icmp unreachable rate-limit 1 burst-size 1

no asdm history enable

arp timeout 14400

timeout xlate 3:00:00

timeout pat-xlate 0:00:30

timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 icmp 0:00:02

timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00

timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00

timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute

timeout tcp-proxy-reassembly 0:01:00

timeout floating-conn 0:00:00

user-identity default-domain LOCAL

http server enable

```
http 192.168.99.0 255.255.255.0 Manage
no snmp-server location
no snmp-server contact
snmp-server enable traps snmp authentication linkup linkdown coldstart warmstart
crypto ipsec security-association pmtu-aging infinite
telnet timeout 5
ssh timeout 5
no threat-detection statistics tcp-intercept
!
class-map inspection_default
  match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
  parameters
    message-length maximum client auto
    message-length maximum 512
policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect h323 h225
    inspect h323 ras
    inspect ip-options
    inspect netbios
    inspect rsh
    inspect rtsp
    inspect skinny
    inspect esmtp
    inspect sqlnet
    inspect sunrpc
    inspect tftp
```

```
inspect sip
inspect xdmcp
!
service-policy global_policy global
Cryptochecksum:910d9cfd63b55877c4359dfbfd2c6acd
: end
```

7.2 Admin secondary

```
: Saved
:
ASA Version 9.1(5) <context>
!
hostname Admin
enable password 8Ry2YjIyt7RRXU24 encrypted
xlate per-session deny tcp any4 any4
xlate per-session deny tcp any4 any6
xlate per-session deny tcp any6 any4
xlate per-session deny tcp any6 any6
xlate per-session deny udp any4 any4 eq domain
xlate per-session deny udp any4 any6 eq domain
xlate per-session deny udp any6 any4 eq domain
xlate per-session deny udp any6 any6 eq domain
names
!
interface Mange
management-only
nameif Mange
security-level 100
ip address 192.168.99.1 255.255.255.0
!
pager lines 24
mtu Mange 1500
icmp unreachable rate-limit 1 burst-size 1
```

no asdm history enable
arp timeout 14400
timeout xlate 3:00:00
timeout pat-xlate 0:00:30
timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 icmp 0:00:02
timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00
timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00
timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute
timeout tcp-proxy-reassembly 0:01:00
timeout floating-conn 0:00:00
user-identity default-domain LOCAL
http server enable
http 192.168.99.0 255.255.255.0 Mange
no snmp-server location
no snmp-server contact
snmp-server enable traps snmp authentication linkup linkdown coldstart warmstart
crypto ipsec security-association pmtu-aging infinite
telnet timeout 5
ssh stricthostkeycheck
ssh timeout 5
ssh key-exchange group dh-group1-sha1
no threat-detection statistics tcp-intercept
!
class-map inspection_default
match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
parameters
message-length maximum client auto
message-length maximum 512
policy-map global_policy

```
class inspection_default
inspect dns preset_dns_map
inspect ftp
inspect h323 h225
inspect h323 ras
inspect ip-options
inspect netbios
inspect rsh
inspect rtsp
inspect skinny
inspect esmtp
inspect sqlnet
inspect sunrpc
inspect tftp
inspect sip
inspect xdmcp
!
service-policy global_policy global
Cryptochecksum:367083b6c713ff8c166f6ca5d2ddb7f0
: end
```

7.3 A-Ward primary

```
: Saved
:
ASA Version 9.0(3) <context>
!
hostname A-Ward
enable password 8Ry2YjIyt7RRXU24 encrypted
names
!
interface Ainside
nameif INSIDE
security-level 100
```



```
ip address 10.0.10.2 255.0.0.0 standby 10.0.10.3
asr-group 1
!
interface Aoutside
nameif OUTSIDE
security-level 0
ip address 172.16.10.2 255.255.0.0 standby 172.16.10.3
asr-group 2
!
pager lines 24
mtu INSIDE 1500
mtu OUTSIDE 1500
monitor-interface INSIDE
monitor-interface OUTSIDE
icmp unreachable rate-limit 1 burst-size 1
no asdm history enable
arp timeout 14400
route OUTSIDE 0.0.0.0 0.0.0.0 172.16.10.1 1
route INSIDE 192.168.10.0 255.255.255.0 10.0.10.1 1
route INSIDE 192.168.20.0 255.255.255.0 10.0.20.1 1
timeout xlate 3:00:00
timeout pat-xlate 0:00:30
timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 icmp 0:00:02
timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00
timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00
timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute
timeout tcp-proxy-reassembly 0:01:00
timeout floating-conn 0:00:00
user-identity default-domain LOCAL
http server enable
http 192.168.99.0 255.255.255.0 INSIDE
no snmp-server location
```

```
no snmp-server contact
snmp-server enable traps snmp authentication linkup linkdown coldstart warmstart
crypto ipsec security-association pmtu-aging infinite
telnet timeout 5
ssh timeout 5
no threat-detection statistics tcp-intercept
!
class-map inspection_default
  match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
  parameters
    message-length maximum client auto
    message-length maximum 512
policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect h323 h225
    inspect h323 ras
    inspect ip-options
    inspect netbios
    inspect rsh
    inspect rtsp
    inspect skinny
    inspect esmtp
    inspect sqlnet
    inspect sunrpc
    inspect tftp
    inspect sip
    inspect xdmcp
```

```
inspect icmp
!
service-policy global_policy global
Cryptochecksum:97297e210af14aa11e95d6a5f2adddb
: end
```

7.4 A-Ward secondary

: Saved

:

ASA Version 9.1(5) <context>

!

hostname A-Ward

enable password 8Ry2YjIyt7RRXU24 encrypted

xlate per-session deny tcp any4 any4

xlate per-session deny tcp any4 any6

xlate per-session deny tcp any6 any4

xlate per-session deny tcp any6 any6

xlate per-session deny udp any4 any4 eq domain

xlate per-session deny udp any4 any6 eq domain

xlate per-session deny udp any6 any4 eq domain

xlate per-session deny udp any6 any6 eq domain

names

!

interface Aoutside

nameif OUTSIDE

security-level 0

ip address 172.16.10.2 255.255.0.0 standby 172.16.10.3

asr-group 2

!

interface Ainside

nameif INSIDE

security-level 100

```
ip address 10.0.10.2 255.0.0.0 standby 10.0.10.3
asr-group 1
!
pager lines 24
mtu INSIDE 1500
mtu OUTSIDE 1500
monitor-interface INSIDE
monitor-interface OUTSIDE
icmp unreachable rate-limit 1 burst-size 1
no asdm history enable
arp timeout 14400
route OUTSIDE 0.0.0.0 0.0.0.0 172.16.10.1 1
route INSIDE 192.168.10.0 255.255.255.0 10.0.10.1 1
route INSIDE 192.168.20.0 255.255.255.0 10.0.20.1 1
timeout xlate 3:00:00
timeout pat-xlate 0:00:30
timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 icmp 0:00:02
timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00
timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00
timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute
timeout tcp-proxy-reassembly 0:01:00
timeout floating-conn 0:00:00
user-identity default-domain LOCAL
http server enable
http 192.168.99.0 255.255.255.0 INSIDE
no snmp-server location
no snmp-server contact
snmp-server enable traps snmp authentication linkup linkdown coldstart warmstart
crypto ipsec security-association pmtu-aging infinite
telnet timeout 5
ssh stricthostkeycheck
ssh timeout 5
```

```
ssh key-exchange group dh-group1-sha1
no threat-detection statistics tcp-intercept
!
class-map inspection_default
  match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
  parameters
    message-length maximum client auto
    message-length maximum 512
policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect h323 h225
    inspect h323 ras
    inspect ip-options
    inspect netbios
    inspect rsh
    inspect rtsp
    inspect skinny
    inspect esmtp
    inspect sqlnet
    inspect sunrpc
    inspect tftp
    inspect sip
    inspect xdmcp
    inspect icmp
!
service-policy global_policy global
Cryptochecksum:6550106db985a9259229597555aaa69a
```

: end

7.5 B-Ward primary

: Saved

:

ASA Version 9.0(3) <context>

!

hostname B-Ward

enable password 8Ry2YjIyt7RRXU24 encrypted

names

!

interface Binside

nameif inside

security-level 100

ip address 10.0.20.2 255.255.255.0 standby 10.0.20.3

asr-group 1

!

interface Boutside

nameif outside

security-level 0

ip address 172.16.20.2 255.255.255.0 standby 172.16.20.3

asr-group 2

!

pager lines 24

mtu outside 1500

mtu inside 1500

monitor-interface outside

monitor-interface inside

icmp unreachable rate-limit 1 burst-size 1

no asdm history enable

arp timeout 14400

route outside 0.0.0.0 0.0.0.0 172.16.20.1 1

route inside 192.168.10.0 255.255.255.0 10.0.10.1 1

```
route inside 192.168.20.0 255.255.255.0 10.0.20.1 1
timeout xlate 3:00:00
timeout pat-xlate 0:00:30
timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 icmp 0:00:02
timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00
timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00
timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute
timeout tcp-proxy-reassembly 0:01:00
timeout floating-conn 0:00:00
user-identity default-domain LOCAL
no snmp-server location
no snmp-server contact
snmp-server enable traps snmp authentication linkup linkdown coldstart warmstart
crypto ipsec security-association pmtu-aging infinite
telnet timeout 5
ssh timeout 5
no threat-detection statistics tcp-intercept
!
class-map inspection_default
  match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
  parameters
    message-length maximum client auto
    message-length maximum 512
policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect h323 h225
    inspect h323 ras
```



```
inspect ip-options
inspect netbios
inspect rsh
inspect rtsp
inspect skinny
inspect esmtp
inspect sqlnet
inspect sunrpc
inspect tftp
inspect sip
inspect xdmcp
inspect icmp
!
service-policy global_policy global
Cryptochecksum:c0060fdec235e80738c15045065f2d9a
: end
```

7.6 B-Ward secondary

```
: Saved
:
ASA Version 9.1(5) <context>
!
hostname B-Ward
enable password 8Ry2YjIyt7RRXU24 encrypted
xlate per-session deny tcp any4 any4
xlate per-session deny tcp any4 any6
xlate per-session deny tcp any6 any4
xlate per-session deny tcp any6 any6
xlate per-session deny udp any4 any4 eq domain
xlate per-session deny udp any4 any6 eq domain
xlate per-session deny udp any6 any4 eq domain
xlate per-session deny udp any6 any6 eq domain
names
```

!

interface Boutside

nameif outside

security-level 0

ip address 172.16.20.2 255.255.255.0 standby 172.16.20.3

asr-group 2

!

interface Binside

nameif inside

security-level 100

ip address 10.0.20.2 255.255.255.0 standby 10.0.20.3

asr-group 1

!

pager lines 24

mtu inside 1500

mtu outside 1500

monitor-interface inside

monitor-interface outside

icmp unreachable rate-limit 1 burst-size 1

no asdm history enable

arp timeout 14400

route outside 0.0.0.0 0.0.0.0 172.16.20.1 1

route inside 192.168.10.0 255.255.255.0 10.0.10.1 1

route inside 192.168.20.0 255.255.255.0 10.0.20.1 1

timeout xlate 3:00:00

timeout pat-xlate 0:00:30

timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 icmp 0:00:02

timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00

timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00

timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute

timeout tcp-proxy-reassembly 0:01:00

timeout floating-conn 0:00:00

```
user-identity default-domain LOCAL
no snmp-server location
no snmp-server contact
snmp-server enable traps snmp authentication linkup linkdown coldstart warmstart
crypto ipsec security-association pmtu-aging infinite
telnet timeout 5
ssh stricthostkeycheck
ssh timeout 5
ssh key-exchange group dh-group1-sha1
no threat-detection statistics tcp-intercept
!
class-map inspection_default
  match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
  parameters
    message-length maximum client auto
    message-length maximum 512
policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect h323 h225
    inspect h323 ras
    inspect ip-options
    inspect netbios
    inspect rsh
    inspect rtsp
    inspect skinny
    inspect esmtp
    inspect sqlnet
```

```
inspect sunrpc
inspect tftp
inspect sip
inspect xdmcp
inspect icmp
!
service-policy global_policy global
Cryptochecksum:cfb60ca4cd346c4d49fe38418aa9edaf
: end
```

7.7 EDGEs

Current configuration : 5568 bytes

```
!
! Last configuration change at 14:55:05 UTC Fri Dec 1 2023
!
version 16.6
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core
!
hostname EDGEs
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
address-family ipv4
exit-address-family
!
```

address-family ipv6

exit-address-family

!

enable secret 5 \$1\$6Nh0\$GZ.rLddji7qnofXDMUc/s.

!

no aaa new-model

clock timezone UTC 1 0

!

no ip domain lookup

ip domain name pokaltagarna.se

!

!

!

!

!

!

!

!

!

!

subscriber templating

!

!

!

!

!

!

!

multilink bundle-name authenticated

!

!

!

!
!
!

crypto pki trustpoint TP-self-signed-2170226186

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-2170226186

revocation-check none

rsakeypair TP-self-signed-2170226186

!
!

crypto pki certificate chain TP-self-signed-2170226186

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 32313730 32323631 3836301E 170D3133 30313130 31343232
30375A17 0D323030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 31373032
32363138 36308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 0100C8E6 D6F6AA8B 44CCA15B 6113ACAB 6710D93D DAA4329D 9F3D2EDC
7817B819 8F29F522 5C774E90 6FC1E98B 83581DD1 2E635BF5 630596AA 281DD611
3D0435E9 5FE84B41 C500BD41 A858E61F E31D28DB B0814D96 079A0C59 347C7845
95BA4975 35EA3319 E84279FB BB251CC5 7760535A 751C9841 5FA31A3A A447E350
9003E693 4D49B7A1 210B7C93 10393182 FBB9FA3D 3180A781 170E03F9 0C41ADC2
FDAC9BD3 F2EBB6DF 1AFD45B0 5D65C026 9EEA3C48 4688D984 498A58CC 52589653
1E993DDA 6D141171 85D9994B 7A75266A 6AE58EFC 2B9276F2 878D5D63 5E60C515
4F8CF26A 3D82E767 88F725B6 01E558F7 F3133AEC 4ED2212B BAC97BF7 7ACF02CD
CB93746C CF7B0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 1470663D F81D1F19 C634476F 62AE4BB3 AEC66F56
A7301D06 03551D0E 04160414 70663DF8 1D1F19C6 34476F62 AE4BB3AE C66F56A7
300D0609 2A864886 F70D0101 05050003 82010100 85BA3F45 6BDEE84C 9854E44B
AEEAF859 51AC261A 39085C6B CBF508CE BB00C063 8465F8B6 EDCDC4F9 A61EDB85
8E60B2DD D9EC1114 53E81199 3A3AA7F0 71075EA4 3FCC7814 CA035E16 88BD9A94

440A8B8A A39AA7B7 4F8B9143 D4991507 111C7F96 0C6109E3 CFB5E3FD 8B4667E2
8538EF77 93671DE2 031288D8 86AB58BA 53972457 B9E04AB8 C991C9D2 F31B3F58
DE4409FD 5C11EBFC B3EEA364 3423F8DF C4DCDED2 04309345 7C3F1303 F41E0CF1
52BDFF1C 88FDBB88 4DFB4B0B C1A53E5A C6D1AB2B B3B45A5C 0BC2BBDD 5A0E707B
31D10FF3 F9069D73 3264FF5E F91250DB 918B8F73 41F0FF66 304C8427 F412101D
3E6AA8E4 826CA0C8 73702B59 EEC073BF ABFD6B13

quit

!

!

!

!

!

!

!

!

!

license udi pid ISR4321/K9 sn FDO24380JVN

license boot suite AdvUCSuiteK9

license boot level securityk9

diagnostic bootup level minimal

spanning-tree extend system-id

!

!

!

username admin privilege 15 secret 9

\$9\$bVTjZveDiWaBt7\$vWe26o/tW.2wsrk7SMK9Pn5BHkQEY3tWRFh3LHoRT2Q

!

redundancy

mode none

!

!

!

!

!

!

track 4 ip sla 4 reachability

delay down 10 up 15

!

track 8 ip sla 8 reachability

delay down 10 up 15

!

!

!

!

!

!

!

!

!

!

!

!

!

!

1

1

•

```
interface GigabitEthernet0/0/0
```

```
ip address 10.0.0.1 255.255.255.0
```

negotiation auto

!

```
interface GigabitEthernet0/0/0.15
```

encapsulation dot1Q 15


```
ip address 172.16.10.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/0.25
encapsulation dot1Q 25
ip address 172.16.20.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/1
ip address dhcp
ip nat outside
negotiation auto
!
interface Serial0/1/0
no ip address
!
interface Serial0/1/1
no ip address
!
interface Serial0/2/0
no ip address
!
interface Serial0/2/1
no ip address
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
ip nat inside source list ALLOWED_PAT interface GigabitEthernet0/0/1 overload
```

```
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip http client source-interface GigabitEthernet0/0/1
ip route 192.168.10.0 255.255.255.0 172.16.10.2 track 4
ip route 192.168.20.0 255.255.255.0 172.16.20.2 track 8
ip route 192.168.0.0 255.255.0.0 172.16.10.2
ip route 192.168.10.0 255.255.255.0 172.16.10.2
ip route 192.168.10.0 255.255.255.0 172.16.10.3 200
ip route 192.168.20.0 255.255.255.0 172.16.20.3 200
!
ip ssh version 2
!
!
ip access-list standard ALLOWED_PAT
permit 192.168.0.0 0.0.255.255
ip sla 4
icmp-echo 172.16.10.2
frequency 5
ip sla schedule 4 life forever start-time now
ip sla 8
icmp-echo 172.16.20.2
frequency 5
ip sla schedule 8 life forever start-time now
!
!
!
!
control-plane
!
!
```

```
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
!
banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C
!
line con 0
exec-timeout 0 0
password 7 05080F1C2243
logging synchronous
login
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
exec-timeout 0 0
logging synchronous
login local
transport input ssh
line vty 5 15
exec-timeout 0 0
logging synchronous
login local
transport input ssh
```

```
!  
ntp server 216.239.35.0  
ntp server 34.208.249.133  
wsma agent exec  
!  
wsma agent config  
!  
wsma agent filesystem  
!  
wsma agent notify  
!  
!  
end
```

7.8 HQ

Building configuration...

```
Current configuration : 5976 bytes  
!  
! Last configuration change at 16:13:35 UTC Sat Jan 12 2013  
!  
version 16.6  
service timestamps debug datetime msec  
service timestamps log datetime msec  
service password-encryption  
platform qfp utilization monitor load 80  
no platform punt-keepalive disable-kernel-core  
!  
hostname HQ  
!  
boot-start-marker  
boot-end-marker
```

!

!

vrf definition A-Ward

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

vrf definition B-Ward

!

address-family ipv4

exit-address-family

!

vrf definition Mgmt-intf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

enable secret 5 \$1\$LmBk\$x24hbfhJwhiUw/xFvYV7P0

!

no aaa new-model

clock timezone UTC 1 0

!

!

!

!

!

!

!

no ip domain lookup

ip domain name pokaltagarna.se

!

!

!

!

!

!

!

!

!

!

subscriber templating

!

!

!

!

!

!

!

multilink bundle-name authenticated

!

!

!

!

crypto pki trustpoint TP-self-signed-2345577025

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-2345577025

revocation-check none

rsakeypair TP-self-signed-2345577025

!

crypto pki trustpoint TP-self-signed-2170226186

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-2170226186

revocation-check none

rsakeypair TP-self-signed-2170226186

!

crypto pki trustpoint TP-self-signed-263984172

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-263984172

revocation-check none

rsakeypair TP-self-signed-263984172

!

!

crypto pki certificate chain TP-self-signed-2345577025

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 32333435 35373730 3235301E 170D3133 30313132 30383438
33345A17 0D323030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 33343535
37373032 35308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 010090E6 EE6C53C3 62DA949F 96A42466 2DF340E9 0050A174 0616BB1F
1F8764B0 79F13606 D7AB28AB FBC35DB9 4E57DEE1 D22D450F B01709DA 15ABE81E
09B0E7A1 ED61032A 5741323C 3D32B0E3 4D699A5B 6AD9562B 0FFFFF06A 4D7180CF
B50096A3 949F798D B82992E0 89078B94 F1A5814F B4708D17 097E69AF 9D09502F
B0C198C8 2B055A07 B328B1CE 0BE0A564 BAEA1297 50C17108 5B293D8B E9C2B130
2C69599F C348DB21 4FA4E625 8276F1DF E6B46B08 FC72C893 4877A543 72EE8982
815FA6FB D884637C CB3EB39A 2905677D 70086C71 5B13CE44 2DADD922 032B396A
217E3CBB CE9ADCBF B0E55DB9 3E89B000 DB544414 01993780 2F6661A8 5AC11F1F
B20D7D5C 50430203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 14751831 875BC029 C1AFB600 92DC7942 07424EBD

AB301D06 03551D0E 04160414 75183187 5BC029C1 AFB60092 DC794207 424EBDAB
300D0609 2A864886 F70D0101 05050003 82010100 5CE3EEAA B8A6AE9B 1D0AC55C
C1E1F986 ED9676E8 4BB7C41D 97CEA5D2 B32C2BB4 73BA41DC 243AFEC5 EB0122BA
A9965DA5 2ED75128 AD787C42 2E8CCBAE 0895E1DA 2BCCB8E9 D634ED92 E1C51A77
EC4B4490 79B9B4F2 E92D3D57 427EC330 E3D57476 20D6DD2C 7DB9CE10 EF6E1C2F
8E8304D8 92ED6B11 979CB0AF 7D6EB57E 204D52E0 FEA6A68C A6E0C313 BBE53735
3226803E 2DA82490 8D8E3D56 2856434C 45AD5CFC ECCDE266 BDEBFDBE C68B44F9
8DA1498D 1A752475 08C8FB87 B190C4B5 5DADC3FB F3442042 8929FA27 C1425FF7
805715FE EE7DCF20 A1A0CCE1 D658FF8D DA7A6B58 083B4927 C439D096 144289ED
D9D6FB29 8BCBE3F0 68AB254E F7AF5722 09FEDA20

quit

crypto pki certificate chain TP-self-signed-2170226186

crypto pki certificate chain TP-self-signed-263984172

!

!

license udi pid ISR4321/K9 sn FD024380JVD

license boot suite FoundationSuiteK9

diagnostic bootup level minimal

spanning-tree extend system-id

!

!

!

username admin privilege 15 secret 9

\$9\$o0rsT.Zge7LtMo\$D7qXwnIfV6OHrX/dDUmE9bL4IMI2Hp02ou62YqBKqaQ

!

redundancy

mode none

!

!

!

!

!

!

track 4 ip sla 4 reachability

delay down 10 up 15

!

track 8 ip sla 8 reachability

delay down 10 up 15

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

interface GigabitEthernet0/0/0

no ip address

negotiation auto

!

interface GigabitEthernet0/0/0.10

encapsulation dot1Q 10

vrf forwarding A-Ward

```
ip address 10.0.10.1 255.255.255.0
!
interface GigabitEthernet0/0/0.20
encapsulation dot1Q 20
vrf forwarding B-Ward
ip address 10.0.20.1 255.255.255.0
!
interface GigabitEthernet0/0/1
no ip address
negotiation auto
!
interface GigabitEthernet0/0/1.10
encapsulation dot1Q 10
vrf forwarding A-Ward
ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0/1.20
encapsulation dot1Q 20
vrf forwarding B-Ward
ip address 192.168.20.1 255.255.255.0
!
interface Serial0/1/0
no ip address
shutdown
!
interface Serial0/1/1
no ip address
shutdown
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
```

```
shutdown
negotiation auto
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ip route vrf A-Ward 0.0.0.0 0.0.0.0 10.0.10.2 track 4
ip route vrf B-Ward 0.0.0.0 0.0.0.0 10.0.20.2 track 8
ip route vrf A-Ward 0.0.0.0 0.0.0.0 10.0.10.2
ip route vrf A-Ward 0.0.0.0 0.0.0.0 10.0.10.3 5
ip route vrf B-Ward 0.0.0.0 0.0.0.0 10.0.10.3
ip route vrf B-Ward 0.0.0.0 0.0.0.0 10.0.20.3 5
!
ip ssh version 2
!
ip sla 4
icmp-echo 10.0.10.2
vrf A-Ward
frequency 5
ip sla schedule 4 life forever start-time now
ip sla 8
icmp-echo 10.0.20.2
vrf B-Ward
frequency 5
ip sla schedule 8 life forever start-time now
!
!
!
!
control-plane
```

!

banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C

!

line con 0

exec-timeout 0 0

password 7 1511021F0725

logging synchronous

login

transport input none

stopbits 1

line aux 0

stopbits 1

line vty 0 4

exec-timeout 0 0

logging synchronous

login local

transport input ssh

line vty 5 15

exec-timeout 0 0

logging synchronous

login local

transport input ssh

!

wsma agent exec

!

wsma agent config

!

wsma agent filesys

!

wsma agent notify

!

!

end

7.9 Lower-Switch

Building configuration...

Current configuration : 4783 bytes

!

! Last configuration change at 08:08:05 UTC Fri Dec 1 2023

!

version 15.2

no service pad

service timestamps debug datetime msec

service timestamps log datetime msec

service password-encryption

!

hostname Lower-Switch

!

boot-start-marker

boot-end-marker

!

enable secret 9 \$9\$rIOgp3j.ntGvnn\$vh0KJC3leckqWNq3nuC8J9DxMfcmKiK0qyw5.x8xXAs

!

username admin privilege 15 secret 9

\$9\$xQeJa9FeqhjXPs\$qRpz98obTHjKPQCdUXToPO8.57Km5PKF663mBhbAS7o

no aaa new-model

switch 1 provision c1000-48t-4g-l

system mtu routing 1500

!

!

no ip domain-lookup

ip domain-name pokaltagarna.se

!

!

!

!
!
!

crypto pki trustpoint TP-self-signed-1931157120
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1931157120
revocation-check none
rsakeypair TP-self-signed-1931157120

!
!

crypto pki certificate chain TP-self-signed-1931157120

certificate self-signed 01

3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 31393331 31353731 3230301E 170D3233 31323031 30383038
30335A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 39333131
35373132 3030819F 300D0609 2A864886 F70D0101 01050003 818D0030 81890281
8100B2BD 4BBC6557 C0900245 9EA2BD7D 792E5F00 C1D62922 B392137E 322BF6D2
158D0D8D 68760B27 676D8C14 3D907595 D3191C33 91ED3170 33862E81 FCB30315
8602B6A2 64C01B2A 8D38ABF5 415A8DA2 6510FC50 E0BBBD48 83CD7AB5 10B916FC
73C67183 099FB625 CE8200C0 4D2EFFD5 F21AF193 6B46F17D 202D3B04 439E790F
60650203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF 301F0603
551D2304 18301680 140D79F2 8C06E4B6 A6E016FD DDA608E6 BC3055BD 7F301D06
03551D0E 04160414 0D79F28C 06E4B6A6 E016FDDD A608E6BC 3055BD7F 300D0609
2A864886 F70D0101 05050003 81810088 61E73743 155A3D21 40602698 C5E336CF
ED4F7388 9794202F 862AE84C 2B18CAA0 EEF528E1 878D70F6 7E946230 8D130B6D
567F7BAA C56D8432 6FE69DEF 32360F17 F96EDC9A E045EBEE 210D42EA 1F3F0806
33F4235A A4718491 0B635C59 464E9291 447F6B5A 13FF04B0 E5786DD3 7B0BA299
072BF75A E5206E96 7CB45BCE 81E88B

quit

!

```
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
!
!
!
!
!
interface GigabitEthernet1/0/1
switchport access vlan 10
switchport mode access
!
interface GigabitEthernet1/0/2
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
```

```
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
switchport trunk allowed vlan 10,20
switchport mode trunk
!
```



```
interface GigabitEthernet1/0/25
!
interface GigabitEthernet1/0/26
!
interface GigabitEthernet1/0/27
!
interface GigabitEthernet1/0/28
!
interface GigabitEthernet1/0/29
!
interface GigabitEthernet1/0/30
!
interface GigabitEthernet1/0/31
!
interface GigabitEthernet1/0/32
!
interface GigabitEthernet1/0/33
!
interface GigabitEthernet1/0/34
!
interface GigabitEthernet1/0/35
!
interface GigabitEthernet1/0/36
!
interface GigabitEthernet1/0/37
!
interface GigabitEthernet1/0/38
!
interface GigabitEthernet1/0/39
!
interface GigabitEthernet1/0/40
!
```

```
interface GigabitEthernet1/0/41
!
interface GigabitEthernet1/0/42
!
interface GigabitEthernet1/0/43
!
interface GigabitEthernet1/0/44
!
interface GigabitEthernet1/0/45
!
interface GigabitEthernet1/0/46
!
interface GigabitEthernet1/0/47
!
interface GigabitEthernet1/0/48
!
interface GigabitEthernet1/0/49
!
interface GigabitEthernet1/0/50
!
interface GigabitEthernet1/0/51
!
interface GigabitEthernet1/0/52
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
ip address 192.168.10.10 255.255.255.0
!
interface Vlan20
```

```
ip address 192.168.20.10 255.255.255.0
```

```
!
```

```
ip http server
```

```
ip http secure-server
```

```
ip ssh version 2
```

```
!
```

```
!
```

```
!
```

```
banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C
```

```
!
```

```
line con 0
```

```
exec-timeout 0 0
```

```
password 7 030752180500
```

```
logging synchronous
```

```
login
```

```
line vty 0 4
```

```
exec-timeout 0 0
```

```
logging synchronous
```

```
login local
```

```
transport input ssh
```

```
line vty 5 15
```

```
exec-timeout 0 0
```

```
logging synchronous
```

```
login local
```

```
transport input ssh
```

```
!
```

```
end
```

7.10 Middle-Switch

Building configuration...

Current configuration : 4786 bytes

```
!
```

! Last configuration change at 09:06:12 UTC Fri Dec 1 2023

!

version 15.2

no service pad

service timestamps debug datetime msec

service timestamps log datetime msec

service password-encryption

!

hostname Middle-Switch

!

boot-start-marker

boot-end-marker

!

enable secret 9 \$9\$dgB62GoXthmWHH\$4lY88gjvddNcWb9cJH2p.P200qF/uHF4./QdBGx6cFQ

!

username admin privilege 15 secret 9

\$9\$zLFkKNaT2l5Yw7\$.Y.dz0AmwWtJgyQtTAb3g7KJzhHQjvgzxhc.Fncl9as

no aaa new-model

switch 1 provision c1000-48t-4g-l

system mtu routing 1500

!

!

no ip domain-lookup

ip domain-name pokaltagarna.se

!

!

!

!

!

!

crypto pki trustpoint TP-self-signed-1931128576

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-1931128576

revocation-check none

rsakeypair TP-self-signed-1931128576

!

!

crypto pki certificate chain TP-self-signed-1931128576

certificate self-signed 01

3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 31393331 31323835 3736301E 170D3233 31323031 30383030
30365A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 39333131
32383537 3630819F 300D0609 2A864886 F70D0101 01050003 818D0030 81890281
8100E982 C114C1D0 89764065 52CD13D3 A47EB661 7F9188B8 B232AEBE 3AB9A8BF
F10B59A5 318989ED 3439B2C5 B5466F0C DACAFCB3 93A783DA 6BF88CAA 4EE0C2CE
D3E4FBF0 DE9D7689 C7F0D74C 751ABDF3 7886F684 0F54425E 93AC9346 7847899E
2F524276 8B9BC883 8ACADCBE 4F778D60 70AFB692 1ACCD7C6 4AC1FE97 D341DDF2
D4270203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF 301F0603
551D2304 18301680 14CFE5E7 9605A11A FB5FE121 A7DFFA8E 182BB4E4 02301D06
03551D0E 04160414 CFE5E796 05A11AFB 5FE121A7 DFFA8E18 2BB4E402 300D0609
2A864886 F70D0101 05050003 8181000E A2F2D194 6F649776 91ACD9CC BB5A3881
D3FDE100 F13F3DD5 6F723997 4CE14B67 190CECC0 9C4E3787 411A860B FC1373DB
63D326FF 1B54C845 0F5D023D EE15EFC7 A60BC7F5 AACFF5AD B59F6CC8 79E6F748
8CDE712A FA7270F1 292C37BB 2BFE0A91 C78C8302 B2DA1734 691F7F96 F67D59DC
1EC67654 A5148C5E F6AA8021 1562B8

quit

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

!

vlan internal allocation policy ascending

!

```
!  
!  
!  
!  
!  
interface GigabitEthernet1/0/1  
  switchport trunk allowed vlan 10,20  
  switchport mode trunk  
!  
interface GigabitEthernet1/0/2  
  switchport trunk allowed vlan 10,20  
  switchport mode trunk  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
!  
interface GigabitEthernet1/0/5  
!  
interface GigabitEthernet1/0/6  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12
```

```
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18  
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
!  
interface GigabitEthernet1/0/22  
!  
interface GigabitEthernet1/0/23  
!  
interface GigabitEthernet1/0/24  
    switchport trunk allowed vlan 10,20  
    switchport mode trunk  
!  
interface GigabitEthernet1/0/25  
!  
interface GigabitEthernet1/0/26  
!  
interface GigabitEthernet1/0/27
```

!

interface GigabitEthernet1/0/28

!

interface GigabitEthernet1/0/29

!

interface GigabitEthernet1/0/30

!

interface GigabitEthernet1/0/31

!

interface GigabitEthernet1/0/32

!

interface GigabitEthernet1/0/33

!

interface GigabitEthernet1/0/34

!

interface GigabitEthernet1/0/35

!

interface GigabitEthernet1/0/36

!

interface GigabitEthernet1/0/37

!

interface GigabitEthernet1/0/38

!

interface GigabitEthernet1/0/39

!

interface GigabitEthernet1/0/40

!

interface GigabitEthernet1/0/41

!

interface GigabitEthernet1/0/42

!

interface GigabitEthernet1/0/43


```
!  
interface GigabitEthernet1/0/44  
!  
interface GigabitEthernet1/0/45  
!  
interface GigabitEthernet1/0/46  
!  
interface GigabitEthernet1/0/47  
!  
interface GigabitEthernet1/0/48  
!  
interface GigabitEthernet1/0/49  
!  
interface GigabitEthernet1/0/50  
!  
interface GigabitEthernet1/0/51  
!  
interface GigabitEthernet1/0/52  
!  
interface Vlan1  
no ip address  
!  
interface Vlan10  
ip address 10.0.10.10 255.255.255.0  
!  
interface Vlan20  
ip address 10.0.20.10 255.255.255.0  
!  
ip http server  
ip http secure-server  
ip ssh version 2  
!
```

```
!  
!  
banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C  
!  
line con 0  
exec-timeout 0 0  
password 7 060506324F41  
logging synchronous  
login  
line vty 0 4  
exec-timeout 0 0  
logging synchronous  
login local  
transport input ssh  
line vty 5 15  
exec-timeout 0 0  
logging synchronous  
login local  
transport input ssh  
!  
end
```

7.11 Upper-Switch

Building configuration...

Current configuration : 3892 bytes

```
!  
! Last configuration change at 08:32:20 UTC Fri Dec 1 2023  
!  
version 15.2  
no service pad  
service timestamps debug datetime msec  
service timestamps log datetime msec
```

no service password-encryption

!

hostname Upper-Switch

!

boot-start-marker

boot-end-marker

!

enable secret 9 \$9\$o4YMsZD6Bx/4yG\$wXMBJrAJlYn186LBd5wO52sLQb6takfGw2VF65298ZA

!

username admin privilege 15 secret 9

\$9\$27MFe3SzaNh4/U\$xFZUhY6ikk.momWBwauPjhuEmBSf4F3XTv9vvo1OHJc

no aaa new-model

switch 1 provision ws-c2960x-24ts-l

!

!

!

!

!

!

no ip domain-lookup

ip domain-name pokaltagarna.se

!

!

!

!

!

!

!

crypto pki trustpoint TP-self-signed-3022554624

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-3022554624

revocation-check none

rsa-keypair TP-self-signed-3022554624

!

!

crypto pki certificate chain TP-self-signed-3022554624

certificate self-signed 01

3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 33303232 35353436 3234301E 170D3233 31323031 30383332
32345A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D33 30323235
35343632 3430819F 300D0609 2A864886 F70D0101 01050003 818D0030 81890281
8100BBE2 EC26D7C3 62105538 3461DB45 CCA15E54 ED3D88E4 CE60C6AE 94FF7D5A
1A8A2A2B 7739797A CA35EDB3 0D09B9C5 009CD739 B1062078 435C01C8 60E8913C
718664CE 8272A708 34074B74 9D366950 5A449045 99FD7B04 AA0F2EC6 1CA6503E
EAF51E06 7D57E314 3C441540 A0C11040 4D09FD1C 5F2D8825 BF0DC85C 171DAC4A
EC970203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF 301F0603
551D2304 18301680 14C561CE 8797D604 EE7FF776 48B4E4FE 7FA752B0 94301D06
03551D0E 04160414 C561CE87 97D604EE 7FF77648 B4E4FE7F A752B094 300D0609
2A864886 F70D0101 05050003 8181002F 4565C3E4 42AC5982 194273AC A8CC928C
13F25DDF DA296FC0 F30DCA3C 702E68F3 51C036E4 E3BCFDA4 83B88A0D DC154F69
6FA001E1 421DBCFA 360D9BBA A7F21CE7 E0B31101 6D4BB042 663D9E56 D90305C2
AFEC2032 CE53F52E D15D34F5 DAEEDADC 42839C47 20D63C4A 8BF856BC 620E77F9
55534256 1D22B251 1AFF5755 42CC40

quit

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

!

!

!

!

vlan internal allocation policy ascending

!
!
!
!
!
!
!
!
!
!
!

interface FastEthernet0

no ip address

shutdown

!

interface GigabitEthernet1/0/1

switchport trunk allowed vlan 15,25

switchport mode trunk

!

interface GigabitEthernet1/0/2

switchport trunk allowed vlan 15,25

switchport mode trunk

!

interface GigabitEthernet1/0/3

!

interface GigabitEthernet1/0/4

!

interface GigabitEthernet1/0/5

!

interface GigabitEthernet1/0/6

!

interface GigabitEthernet1/0/7

```
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18  
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
!  
interface GigabitEthernet1/0/22  
!  
interface GigabitEthernet1/0/23
```

```
!  
interface GigabitEthernet1/0/24  
  switchport trunk allowed vlan 15,25  
  switchport mode trunk  
!  
interface GigabitEthernet1/0/25  
!  
interface GigabitEthernet1/0/26  
!  
interface GigabitEthernet1/0/27  
!  
interface GigabitEthernet1/0/28  
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
interface Vlan15  
  ip address 172.16.10.10 255.255.255.0  
!  
interface Vlan25  
  ip address 172.16.20.10 255.255.255.0  
!  
ip default-gateway 172.16.10.1  
!  
ip http server  
ip http secure-server  
ip ssh version 2  
!  
!  
!  
banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C
```

```
!  
line con 0  
  exec-timeout 0 0  
line vty 0 4  
  login  
line vty 5 15  
  login  
!  
!  
end
```

7.12 System primary

```
: Saved  
:  
ASA Version 9.0(3) <system>  
!  
hostname ASA-AAA  
enable password 8Ry2YjIyt7RRXU24 encrypted  
no mac-address auto  
!  
interface Ethernet0/0  
  description LAN Failover Interface  
!  
interface Ethernet0/1  
  description STATE Failover Interface  
!  
interface Ethernet0/2  
!  
interface Ethernet0/2.15  
  description A-Ward  
  vlan 15  
!  
interface Ethernet0/2.25
```



```
description B-Ward
vlan 25
!
interface Ethernet0/3
!
interface Ethernet0/3.10
description A-Ward
vlan 10
!
interface Ethernet0/3.20
description B-Ward
vlan 20
!
interface Management0/0
!
class default
limit-resource All 0
limit-resource Mac-addresses 65535
limit-resource ASDM 5
limit-resource SSH 5
limit-resource Telnet 5
!

ftp mode passive
pager lines 24
failover
failover lan unit primary
failover lan interface FailoverLAN Ethernet0/0
failover polltime unit msec 300 holdtime msec 999
failover replication http
failover link FailoverState Ethernet0/1
failover interface ip FailoverLAN 192.168.100.1 255.255.255.0 standby 192.168.100.2
```

failover interface ip FailoverState 192.168.101.1 255.255.255.0 standby 192.168.101.2

failover group 1

preempt 60

failover group 2

secondary

no asdm history enable

arp timeout 14400

no arp permit-nonconnected

console timeout 0

admin-context Admin

context Admin

allocate-interface Management0/0 Manged

config-url disk0:/admin.cfg

!

context A-Ward

allocate-interface Ethernet0/2.15 Aoutside

allocate-interface Ethernet0/3.10 Ainside

config-url disk0:/A-Ward.cfg

join-failover-group 1

!

context B-Ward

allocate-interface Ethernet0/2.25 Boutside

allocate-interface Ethernet0/3.20 Binside

config-url disk0:/B-Ward.cfg

join-failover-group 2

!

prompt hostname context

no call-home reporting anonymous

```
call-home
profile CiscoTAC-1
no active
destination address http https://tools.cisco.com/its/service/oddce/services/DDCEService
destination address email callhome@cisco.com
destination transport-method http
subscribe-to-alert-group diagnostic
subscribe-to-alert-group environment
subscribe-to-alert-group inventory periodic monthly
subscribe-to-alert-group configuration periodic monthly
subscribe-to-alert-group telemetry periodic daily
Cryptochecksum:7d76824b1c8fe460f325f180ad7c2f98
: end
```

7.13 System secondary

```
: Saved

:
ASA Version 9.1(5) <system>
!
hostname ASA-AAA
enable password 8Ry2Yjlyt7RRXU24 encrypted
no mac-address auto
!
interface Ethernet0/0
description LAN Failover Interface
!
interface Ethernet0/1
description STATE Failover Interface
!
interface Ethernet0/2
!
interface Ethernet0/2.15
description A-Ward
```

```
vlan 15
!
interface Ethernet0/2.25
description B-Ward
vlan 25
!
interface Ethernet0/3
!
interface Ethernet0/3.10
description A-Ward
vlan 10
!
interface Ethernet0/3.20
description B-Ward
vlan 20
!
interface Management0/0
!
class default
limit-resource All 0
limit-resource Mac-addresses 65535
limit-resource ASDM 5
limit-resource SSH 5
limit-resource Telnet 5
!

ftp mode passive
pager lines 24
failover
failover lan unit secondary
failover lan interface FailoverLAN Ethernet0/0
failover polltime unit msec 300 holdtime msec 999
```

failover replication http

failover link FailoverState Ethernet0/1

failover interface ip FailoverLAN 192.168.100.1 255.255.255.0 standby 192.168.100.2

failover interface ip FailoverState 192.168.101.1 255.255.255.0 standby 192.168.101.2

failover group 1

preempt 60

failover group 2

secondary

no asdm history enable

arp timeout 14400

no arp permit-nonconnected

no ssh stricthostkeycheck

console timeout 0

admin-context Admin

context Admin

allocate-interface Management0/0 Mange

config-url disk0:/admin.cfg

!

context A-Ward

allocate-interface Ethernet0/2.15 Aoutside

allocate-interface Ethernet0/3.10 Ainside

config-url disk0:/A-Ward.cfg

join-failover-group 1

!

context B-Ward

allocate-interface Ethernet0/2.25 Boutside

allocate-interface Ethernet0/3.20 Binside

config-url disk0:/B-Ward.cfg

join-failover-group 2

!

prompt hostname context

no call-home reporting anonymous

call-home

profile CiscoTAC-1

no active

destination address http <https://tools.cisco.com/its/service/oddce/services/DDCEService>

destination address email callhome@cisco.com

destination transport-method http

subscribe-to-alert-group diagnostic

subscribe-to-alert-group environment

subscribe-to-alert-group inventory periodic monthly

subscribe-to-alert-group configuration periodic monthly

subscribe-to-alert-group telemetry periodic daily

Cryptochecksum:04cbefef95b341b4d66ffea2f9437355

: end